



The Midnight Mango Thief Adventure

A Super Engineer Storybook

Once upon a time, there was a little boy named Ayaan. His grandfather grew the most gorgeous, big, yellow Alphonso mango trees in their backyard orchard in Ratnagiri. The mangoes were sweet, juicy, and absolutely perfect.

But oh no! Every single night at midnight, a sneaky, mischievous little monkey climbed down the branches and took a giant nibble out of the very best fruits! Ayaan could not stay awake all night long to watch the tree.

"Can we use our science superpowers to build a secret electronic sound alarm trap to catch the thief in the act?" Ayaan asked. Let's help him protect the mangoes!

THE SECRET OF THE ELECTRICITY ANTS

To make our sound alarm work, we need to build a special path called a **circuit**.

Imagine that inside every wire, there are billions of tiny, invisible, energetic **Electricity Ants**! These ants love to run and carry power. But they have one major rule: they will *only* start marching if they have a perfect, unbroken circular track to walk on!

- **The Open Circuit:** If there is a gap or a broken line in the wire, the ants freeze because they cannot jump across! Everything stays completely quiet.
- **The Closed Circuit:** If we connect the lines perfectly, it becomes a complete loop. The ants can march all the way through the wires, and the alarm sings!

CIRCLE TRACKS ARE EVERYWHERE

We see these hidden ant tracks working all around us every single day:

- **Example 1, The Big Metro Train:** Think of the busy Metro trains zooming through the city. If the circular track is entirely built and continuous, the train runs smoothly from station to station. If a section of track is removed for repairs, the loop breaks and the train stops instantly!
- **Example 2, Festival Rice Lights:** During festive celebrations like Diwali or Eid, families hang beautiful twinkling fairy lights along their balconies. If just one tiny wire gets clipped or snapped by a sharp window edge, the entire string of lights goes dark immediately because the ant pathway is open!

MEET THE SOUND MAKER: THE BUZZER

Every alarm system needs a loud mouth, and ours is called the **piezo buzzer**! This clever little part takes the fast running speed of our marching electricity ants and turns it instantly into clear sound waves.

Where have you heard a buzzer today?

- When a delivery person pushes your apartment **doorbell** button and it sings *Ding-Dong*!
- When the **school bell** loudly echoes across the hallways to tell you it's time for recess!
- When the **microwave** loudly beeps to tell you your food is warm and ready to eat!

THE SECRET MAT TRIGGER

How will our circuit know when the monkey arrives? We will hide a secret **pressure switch** under a garden doormat!

Normally, the switch just hangs out in an open state. It holds the wires apart so the circuit stays silent and the ants wait patiently. But the very second the mango thief steps his furry foot down on our mat, his weight forces the hidden metal pads inside to touch! The gap closes instantly, the track is connected, and the ants start to run!

BUILDING THE TRAP STEP BY STEP

Let's put our invention together! First, stretch out your magic fingers and look at the workbench. We have a heavy-duty yellow battery pack and our buzzer. Take the positive **red wire** from the battery, pull it across, and snap it directly onto the **red leg** of our buzzer. Wonderful! The power is locked into position.

Next, let's connect the ground line to finish the loop. Take the **black wire** from the buzzer, run it into our secret mat switch, and connect the other side of the switch back to the **black side** of the battery. The loop is complete, and the trap is set!

TESTING THE ALARM AT MIDNIGHT

It is midnight in the orchard! Let's test our trap to make sure it works before the monkey gets to the tree. Imagine dropping a large, heavy stone right on top of our garden doormat switch. Look! The mat squishes down, the hidden metal pads press together, the gap closes, and the electricity ants complete their march all the way to the buzzer!

BZZZZZZZZ!

It works beautifully! Shabash!

BE A SCIENCE DETECTIVE

Sometimes, real inventors have to solve puzzles when things don't work right. Let's look at two common mysteries:

- **The Upside-Down Wires:** What if the buzzer stays totally quiet? Check the wires! Electricity ants from toy batteries are like cars on a one-way street; they can only march in one specific direction. If the red and black wires are accidentally swapped backwards, the ants get confused and freeze. Flipping them back to the right side fixes the track instantly!
- **The Plastic Block Wall:** What if a plastic toy block gets stuck in the circuit? Materials like plastic and rubber are **insulators**, which means they act like giant brick walls to electricity ants. The ants hit the plastic and cannot move forward! To fix it, we must swap the plastic for a metal safety pin. Metals are **conductors**, which means they act like smooth, beautifully paved open highways where ants love to sprint!

THE SECRET SAFETY OATH

Before we celebrate saving the mangoes, every great scientist must learn the golden rules of safety. Small toy batteries have gentle, friendly power that is perfectly safe for young inventors to build with. But the big wall sockets in your house carry giant, dangerous high-voltage electricity! You must **never ever** touch wall outlets or stick anything into them. Always ask a grown-up before doing any science experiment!

Raise your right hand and say the Safety Pledge aloud:

"I promise to always use safe toy batteries for my inventions. I will never, ever touch big household wall plugs, and I will always ask my parents or teacher first!"

Congratulations! You are now an officially certified Junior Lab Assistant!

VICTORY!

Hooray! Ayaan's beautiful mango tree is perfectly safe now, thanks to your incredible engineering skills! The sneaky monkey heard our loud buzzer alarm go *BZZZZ* and safely ran back into the deep jungle without a single bite of fruit.

Before you close this book, here is your fun **Super Engineer Homework**: Look around your kitchen and living room today and try to spot *one* device that uses a buzzer or makes a beep. Draw a picture of it in your notebook to share next time.

Thank you, and keep inventing!